TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Specific Resistance of a Wire by Metre Bridge

Balance the specimen wire against a standard resistance, then turn resistance into resistivity with the wire's measured length and diameter — the material constant behind every conductor.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 3 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the specific resistance (resistivity) ρ of the material of a given wire. The resistance of a uniform wire depends on its geometry and on one property of the material alone:

$$R = \rho \, L / A \quad \rightarrow \quad \rho = R \, A / L = R \, \pi \, d^2 / (4 \, L)$$

where L is the length, d the diameter and A the cross-sectional area. The experiment therefore has two halves: measure R accurately, and measure L and d accurately. R comes from the metre bridge — a stretched uniform wire of 100 cm whose two segments act as the ratio arms of a Wheatstone bridge. With a standard resistance S in the left gap, the specimen X in the right gap, and the null point at l cm from the left,

$$X = S \, (100 - l) / l$$

(Experiment 2's end corrections α, β can be applied: $X = S(100 - l + \beta)/(l + \alpha)$.) Resistivity is quoted in Ω·m: it is the resistance between opposite faces of a unit cube of the material. Copper ≈ 1.7×10^{-8}, constantan ≈ 4.9×10^{-7}, nichrome ≈ 1.1×10^{-6} Ω·m — a span of nearly a hundred, which is why constantan and nichrome, not copper, make resistance boxes and heater elements: their ρ is high and, just as important, barely changes with temperature.

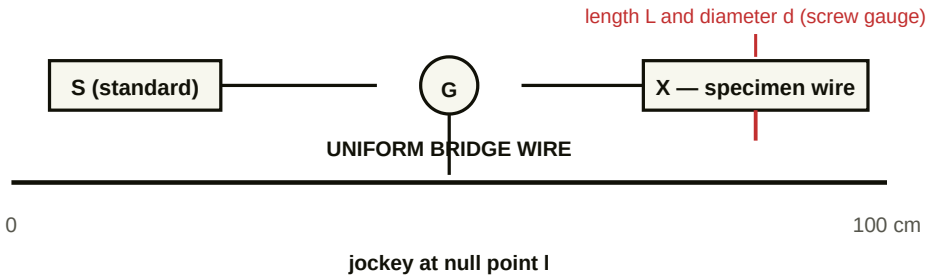


Figure 1 — The specimen wire itself forms the unknown arm. Its resistance comes from the balance; its resistivity needs the geometry — length by rule, diameter by screw gauge at several points and in two perpendicular directions.

2. Apparatus and procedure

- **Apparatus:** metre bridge, specimen wire (about 1 m of constantan or manganin) mounted between terminals, standard resistance box, cell, galvanometer, jockey, key, screw gauge, metre rule.
- Measure the specimen's length L between its terminals with the rule. Measure its diameter with the screw gauge at five or six places along its length, twice at each place in perpendicular directions; average d. (Diameter error is doubled in ρ, since $A \propto d^2$ — this is the measurement that decides the result.)
- Connect S (left gap), specimen (right gap), cell, galvanometer and jockey as in Fig. 1. Set S so the null point falls near the middle (40–60 cm); record l.
- Interchange S and the specimen and record the new null point; computing X from both positions cancels the end corrections to first order. Repeat for two more values of S and average X.
- Compute $\rho = X\pi d^2/(4L)$ in Ω·cm, then convert: 1 Ω·cm = 10^{-2} Ω·m.

3. Worked calculation

Data: S = 4 Ω, null at l = 49.4 cm → X = 4 × (50.6/49.4) = 3.905 Ω ≈ 3.91 Ω. Specimen: L = 100.0 cm; mean diameter d = 0.40 mm = 0.040 cm.

$$A = \pi d^2/4 = \pi(0.040)^2/4 = 1.257 \times 10^{-3}\ \text{cm}^2$$

$\rho = XA/L = (3.905 \times 1.257 \times 10^{-3})/100 = 4.91 \times 10^{-5}$ Ω·cm $= 4.91 \times 10^{-7}$ Ω·m — matching constantan (standard value ≈ 4.9×10^{-7} Ω·m), so the specimen is identified as constantan wire.

4. Precautions

- Keep the bridge current small and brief: heating raises the specimen's resistance during the measurement (constantan's low temperature coefficient is chosen for exactly this reason).
- Measure the diameter with zero-error correction applied, and reject a wire that is kinked or visibly non-uniform — ρ assumes a uniform cylinder.
- Null point in the 40–60 cm window; approach it from both sides; do not drag the jockey.
- Terminals gripping the specimen must bite on clean metal — the measured L is the length between the inner faces of the terminals.

Practice (viva preparation)

1. Define specific resistance in words, without mentioning any formula's symbols first — then give the formula.
2. A wire of length 50 cm and diameter 0.30 mm has resistance 6.0 Ω. Find ρ in Ω·m and guess the material class (conductor, resistance alloy, or heater alloy).
3. Why does a 1 % error in diameter matter more than a 1 % error in length?
4. The same specimen is drawn out to twice its length (volume constant). What happens to its resistance?
5. Why are resistance-box coils made of constantan or manganin rather than copper?

Answer key

1. Specific resistance is the resistance offered between the opposite faces of a unit cube of the material at a stated temperature — a property of the material alone, not of the sample's size. $\rho = RA/L$.
2. $A = \pi(3.0 \times 10^{-4}\ \text{m})^2/4 = 7.07 \times 10^{-8}\ \text{m}^2$. $\rho = RA/L = 6.0 \times 7.07 \times 10^{-8} / 0.50 = 8.5 \times 10^{-7}\ \Omega\cdot\text{m}$ — a resistance alloy (constantan–nichrome range), not a plain conductor.
3. Because $\rho \propto d^2$ but only $\propto 1/L$: a 1 % diameter error becomes a 2 % error in area and hence in ρ, while a 1 % length error stays 1 %. The screw gauge work dominates the accuracy.
4. Doubling L at constant volume halves A, so $R = \rho L/A$ rises by $2 \times 2 = 4$ times: 4R.
5. High ρ (compact coils), a temperature coefficient near zero (the marked value holds when the box warms in use), and stability against oxidation — copper fails all three: low ρ, and a coefficient of about 0.4 % per °C.
